

Course Title : Database Systems Lab

Course Code : CSE 212

Project Report

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Project Description

A. Project Title : Mood-Based Cafe Recommendation System

B. Project Purpose : This project is designed to develop a database system that recommends cafes, menu items, music, and seating arrangements based on a user's mood. The system aims to improve customer experience by providing personalized and comfortable cafe recommendations while efficiently managing bookings, orders, and reviews.

C. Project Overview : The Mood-Based Cafe Recommendation System is a database-based application designed to improve customer experience by recommending cafes according to a user's mood and preferences. Nowadays, people visit cafes not only for food and drinks, but also for relaxation, comfort, entertainment, study, and spending quality time. Because of this, choosing the right cafe environment has become important for many customers.

This project focuses on building a smart recommendation system where users can get cafe suggestions based on their mood and preferred atmosphere. The system stores and manages information related to customers, cafes, moods, menu items, music, seating arrangements, bookings, orders, reviews, and payments. It also maintains proper relationships between different entities using primary keys and foreign keys.

The system helps solve problems such as difficulty in selecting cafes based on mood, lack of personalized recommendations, and poor integration of music, seating, and menu preferences. It also helps manage bookings and customer activities more efficiently. Overall, the system provides a more comfortable and personalized cafe experience for customers while helping cafe owners understand customer preferences better.

D. Objectives :

- Design a relational database system for mood-based cafe recommendations
- Store customer information, preferences, and mood data
- Recommend suitable cafes according to customer mood
- Suggest menu items and music based on preferences
- Manage customer bookings and orders efficiently
- Maintain data integrity using primary keys and foreign keys
- Provide accurate and efficient query-based results

E. System Features :

Functional Requirements:

- Customer can log mood
- System recommends cafe, food, and music
- Customer can book tables
- Customer can place orders
- Payment system is included
- Reviews can be given

Non-Functional Requirements:

- Data consistency ensured
- Referential integrity maintained
- Fast query execution
- Scalable system

Key Technical Features:

- Multiple related tables
- Many-to-many relationships (handled via bridge tables)
- Foreign key constraints
- Composite keys (for recommendation tables)

2(i). Entity–Relationship (ER) Diagram

To create the ER diagram, several entity sets along with their attributes were identified. The diagram visually represents the relationships among these entities, showing how they work together to define the overall structure and functionality of the database.

2(ii). Entity Sets and Attributes

Entity Sets and Attributes

1) Customer

Attributes: Customer_ID, Name, Email, Phone, Gender, Age

2) Mood

Attributes: Mood_ID, Mood_Name, Description

3) Cafe

Attributes: Cafe_ID, Cafe_Name, Rating, Contact_Number

4) Location

Attributes: Location_ID, Area, City

5) Category

Attributes: Category_ID, Category_Name

6) Menu_Item

Attributes: Item_ID, Item_Name, Price

7) Seat_Type

Attributes: SeatType_ID, SeatType_Name

8) Cafe_Table

Attributes: Table_ID, Table_Number, Capacity

9) Music

Attributes: Music_ID, Music_Name, Genre, Artist, Mood_Vibe

10) Customer_Mood_Log

Attributes: Log_ID, Log_Date, Log_Time

11) Booking

Attributes: Booking_ID, Booking_Name, Booking_Time, No_of_People, Status

12) Orders

Attributes: Order_ID, Order_Date, Total_Amount

13) Order_Details

Attributes: OrderDetail_ID, Quantity, Unit_Price

14) Payment

Attributes: Payment_ID, Payment_Date, Amount, Payment_Method, Status, Order_ID

15) Review

Attributes: Review_ID, Rating, Comment, Review_Date

2(iii). Relationships Between Entity Sets

1. Customer – Booking

- A customer can make one or more bookings.
- One booking belongs to one customer.
- **Relationship Type:** One-to-Many

2. Booking – Orders

- One booking can place one or more orders.
- One order belongs to one booking.
- **Relationship Type:** One-to-Many

3. Orders – Payment

- One order has one payment.
- One payment belongs to one order.
- **Relationship Type:** One-to-One

4. Orders – Order_Details

- One order can contain multiple order details.
- One order detail belongs to one order.
- **Relationship Type:** One-to-Many

5. Order_Details – Menu_Item

- One menu item can appear in many order details.
- One order detail contains one menu item.
- **Relationship Type:** One-to-Many

6. Cafe – Menu_Item

- One cafe can offer multiple menu items.
- One menu item belongs to one cafe.
- **Relationship Type:** One-to-Many

7. Cafe – Music

- One cafe can have multiple music items.
- One music item belongs to one cafe.
- **Relationship Type:** One-to-Many

8. Cafe – Cafe_Table

- One cafe can have multiple tables.
- One table belongs to one cafe.
- **Relationship Type:** One-to-Many

9. Seat_Type – Cafe_Table

- One seat type can be assigned to multiple cafe tables.
- One cafe table has one seat type.
- **Relationship Type:** One-to-Many

10. Location – Cafe

- One location can contain multiple cafes.
- One cafe belongs to one location.
- **Relationship Type:** One-to-Many

11. Category – Cafe

- One category can include multiple cafes.
- One cafe belongs to one category.
- **Relationship Type:** One-to-Many

12. Customer – Review

- One customer can write multiple reviews.

- One review is written by one customer.
- **Relationship Type:** One-to-Many

13. Cafe – Review

- One cafe can receive multiple reviews.
- One review belongs to one cafe.
- **Relationship Type:** One-to-Many

14. Customer – Customer_Mood_Log

- One customer can create multiple mood logs.
- One mood log belongs to one customer.
- **Relationship Type:** One-to-Many

15. Mood – Customer_Mood_Log

- One mood can appear in multiple mood logs.
- One mood log refers to one mood.
- **Relationship Type:** One-to-Many

16. Mood – Menu_Item

- One mood can recommend multiple menu items.
- One menu item can be recommended for multiple moods.
- **Relationship Type:** Many-to-Many

17. Mood – Music

- One mood can recommend multiple music items.
- One music item can be recommended for multiple moods.
- **Relationship Type:** Many-to-Many

18. Mood – Seat_Type

- One mood can recommend multiple seat types.
- One seat type can be recommended for multiple moods.
- **Relationship Type:** Many-to-Many

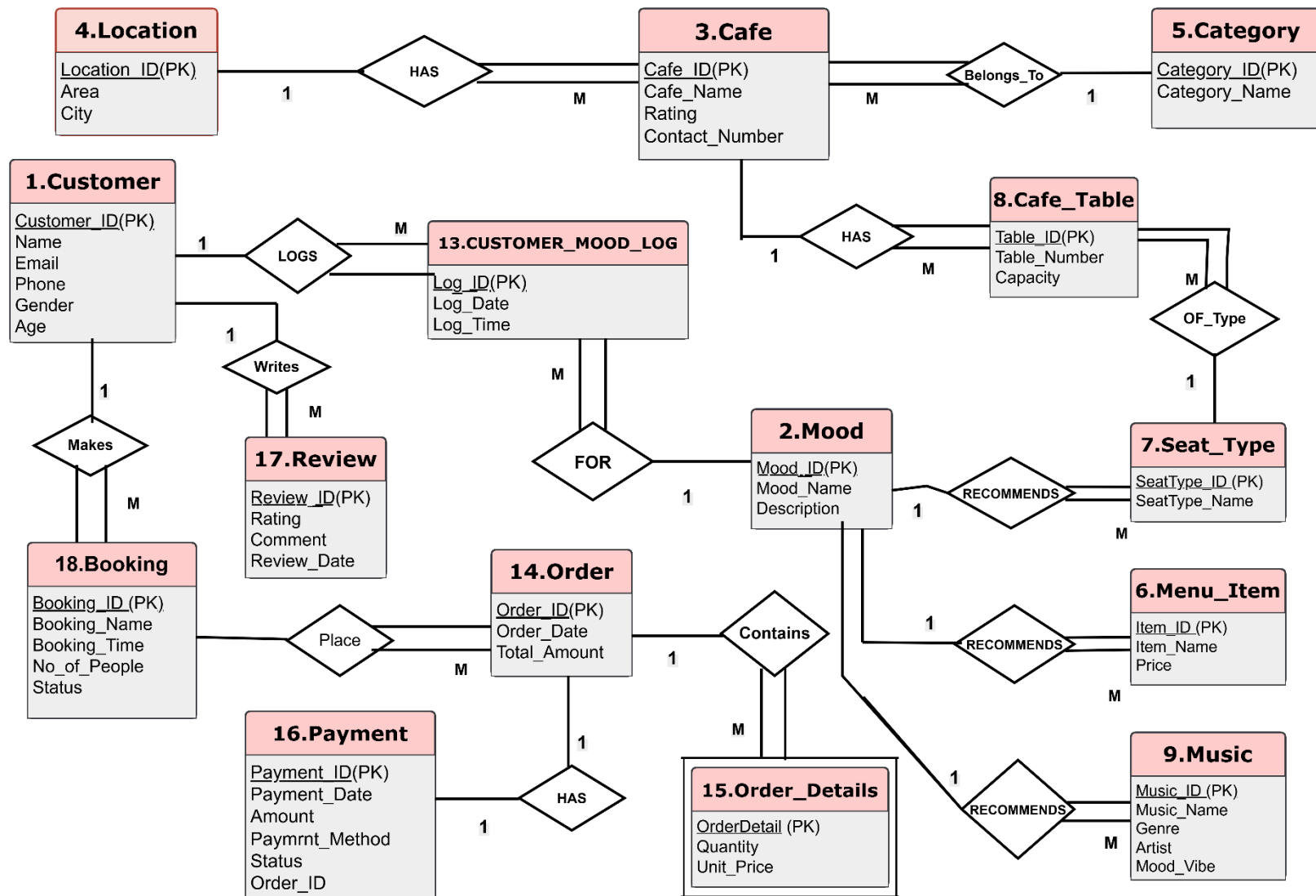
Summary of Relationships :

The Mood-Based Café Recommendation System contains different types of relationships among its entities. Customer, Cafe, Booking, Orders, Payment, Review, Mood, Menu_Item, Music, and Seat_Type are connected through one-to-one, one-to-many, and many-to-many relationships. These relationships help the system manage customer mood logs, café information, bookings, orders, payments, reviews, and mood-based recommendations in an organized way.

The ER diagram on the next page visually represents these relationships and shows how all entity sets are connected within the database system.

2(iv). ER DIAGRAM

MOOD-BASED CAFÉ RECOMMENDATION SYSTEM ER DIAGRAM



3(i) ER to Schema Conversion :

As we have shown the ER diagram of the database, now we prepared a schema table from that ER for better understanding the design implementation with the uses of primary keys and foreign keys.

3(ii) Tables and Attributes :

1. LOCATION

Attributes: Location_ID (PK), Area, City

2. CATEGORY

Attributes: Category_ID (PK), Category_Name

3. CUSTOMER

Attributes: Customer_ID (PK), Name, Email, Phone, Gender, Age

4. MOOD

Attributes: Mood_ID (PK), Mood_Name, Description

5. CAFE

Attributes: Cafe_ID (PK), Cafe_Name, Rating, Contact_Number, Location_ID (FK), Category_ID (FK)

6. MENU_ITEM

Attributes: Item_ID (PK), Item_Name, Price, Cafe_ID (FK)

7. SEAT_TYPE

Attributes: SeatType_ID (PK), SeatType_Name

8. CAFE_TABLE

Attributes: Table_ID (PK), Table_Number, Capacity, Cafe_ID (FK), SeatType_ID (FK)

9. MUSIC

Attributes: Music_ID (PK), Music_Name, Genre, Artist, Mood_Vibe, Cafe_ID (FK)

10. CUSTOMER_MOOD_LOG

Attributes: Log_ID (PK), Log_Date, Log_Time, Customer_ID (FK), Mood_ID (FK), Cafe_ID(FK)

11. BOOKING

Attributes: Booking_ID(PK), Booking_Name, Booking_Time, No_of_People, Status , Customer_ID (FK), Table_ID (FK)

12. ORDERS

Attributes: Order_ID (PK), Order_Date, Total_Amount, Customer_ID (FK), Cafe_ID (FK), Booking_ID (FK)

13. ORDER_DETAILS

Attributes: OrderDetail_ID (PK), Quantity, Unit_Price, Order_ID (FK), Item_ID (FK)

14. PAYMENT

Attributes: Payment_ID (PK), Payment_Date, Amount, Payment_Method, Status, Order_ID (FK)

15. REVIEW

Attributes: Review_ID (PK), Rating, Comment, Review_Date, Customer_ID (FK), Cafe_ID (FK)

16. MOOD_MENU_RECOMMENDATION

Attributes: Mood_ID (PK)(FK), Item_ID (PK)(FK)

17. MOOD_MUSIC_RECOMMENDATION

Attributes: Mood_ID (PK)(FK), Music_ID (PK)(FK)

18. MOOD_SEAT_RECOMMENDATION

Attributes: Mood_ID (PK)(FK), SeatType_ID (PK)(FK)

3(iii). Relation Between tables -Primary keys & Foreign keys :

1. Customer

Description: This table contains all details about the customers who use the Mood-Based Café Recommendation System. It stores customer information such as name, email, phone number, gender, and age. The attribute 'Customer_ID' is the primary key. There are no foreign keys.

2. Mood

Description: This table stores different moods of customers such as happy, relaxed, energetic, or romantic. It also contains a short description of each mood. The attribute 'Mood_ID' is the primary key. There are no foreign keys.

3. Location

Description: This table contains café location information including area and city. The attribute 'Location_ID' is the primary key. There are no foreign keys.

4. Category

Description: This table stores different café categories such as romantic café, study café, or luxury café. The attribute 'Category_ID' is the primary key. There are no foreign keys.

5. Cafe

Description: This table stores all information about cafés registered in the system. It contains café name, rating, and contact number. Each café belongs to a specific location and category. The attribute 'Cafe_ID' is the primary key. The attributes 'Location_ID' and 'Category_ID' are foreign keys referencing the 'Location' and 'Category' tables respectively.

6. Menu_Item

Description: This table stores all menu items offered by cafés. It contains item name and price information. Each menu item belongs to a specific café. The attribute 'Item_ID' is the primary key. The attribute 'Cafe_ID' is a foreign key referencing the 'Cafe' table.

7. Music

Description: This table stores music information played in cafés. It includes music name, genre, artist, and mood vibe. Each music item belongs to a specific café. The attribute 'Music_ID' is the primary key. The attribute 'Cafe_ID' is a foreign key referencing the 'Cafe' table.

8. Seat_Type

Description: This table stores different seating types available in cafés such as couple seat, rooftop seat, or window seat. The attribute 'SeatType_ID' is the primary key. There are no foreign keys.

9. Cafe_Table

Description: This table contains information about café tables and seating capacity. Each table belongs to a specific café and has one seat type. The attribute 'Table_ID' is the primary key. The attributes 'Cafe_ID' and 'SeatType_ID' are foreign keys referencing the 'Cafe' and 'Seat_Type' tables respectively.

10. Customer_Mood_Log

Description: This table stores customer mood activity records. It keeps track of customer moods, selected cafés, and date-time information. The attribute 'Log_ID' is the primary key. The attributes 'Customer_ID', 'Mood_ID', and 'Cafe_ID' are foreign keys referencing the 'Customer', 'Mood', and 'Cafe' tables respectively.

11. Booking

Description: This table stores customer booking information including booking time, number of people, and booking status. Each booking is linked to a customer and a café table. The attribute 'Booking_ID' is the primary key. The attributes 'Customer_ID' and 'Table_ID' are foreign keys referencing the 'Customer' and 'Cafe_Table' tables respectively.

12. Orders

Description: This table records all customer orders placed in cafés. It stores order date and total amount information. Each order is connected to a customer, café, and booking. The attribute

'Order_ID' is the primary key. The attributes 'Customer_ID', 'Cafe_ID', and 'Booking_ID' are foreign keys referencing the 'Customer', 'Cafe', and 'Booking' tables respectively.

13. Order_Details

Description: This table contains detailed information about ordered menu items. It includes quantity and unit price of each item. The attribute 'OrderDetail_ID' is the primary key. The attributes 'Order_ID' and 'Item_ID' are foreign keys referencing the 'Orders' and 'Menu_Item' tables respectively.

14. Payment

Description: This table stores payment information for customer orders. It includes payment amount, method, status, and payment date. The attribute 'Payment_ID' is the primary key. The attribute 'Order_ID' is a foreign key referencing the 'Orders' table.

15. Review

Description: This table stores customer reviews and ratings about cafés. It includes feedback comments and review dates. The attribute 'Review_ID' is the primary key. The attributes 'Customer_ID' and 'Cafe_ID' are foreign keys referencing the 'Customer' and 'Cafe' tables respectively.

16. Mood_Menu_Recommendation

Description: This table stores mood-based menu recommendations. It connects moods with recommended menu items. The attributes 'Mood_ID' and 'Item_ID' together form the primary key and are foreign keys referencing the 'Mood' and 'Menu_Item' tables respectively.

17. Mood_Music_Recommendation

Description: This table stores mood-based music recommendations. It connects moods with recommended music items. The attributes 'Mood_ID' and 'Music_ID' together form the primary key and are foreign keys referencing the 'Mood' and 'Music' tables respectively.

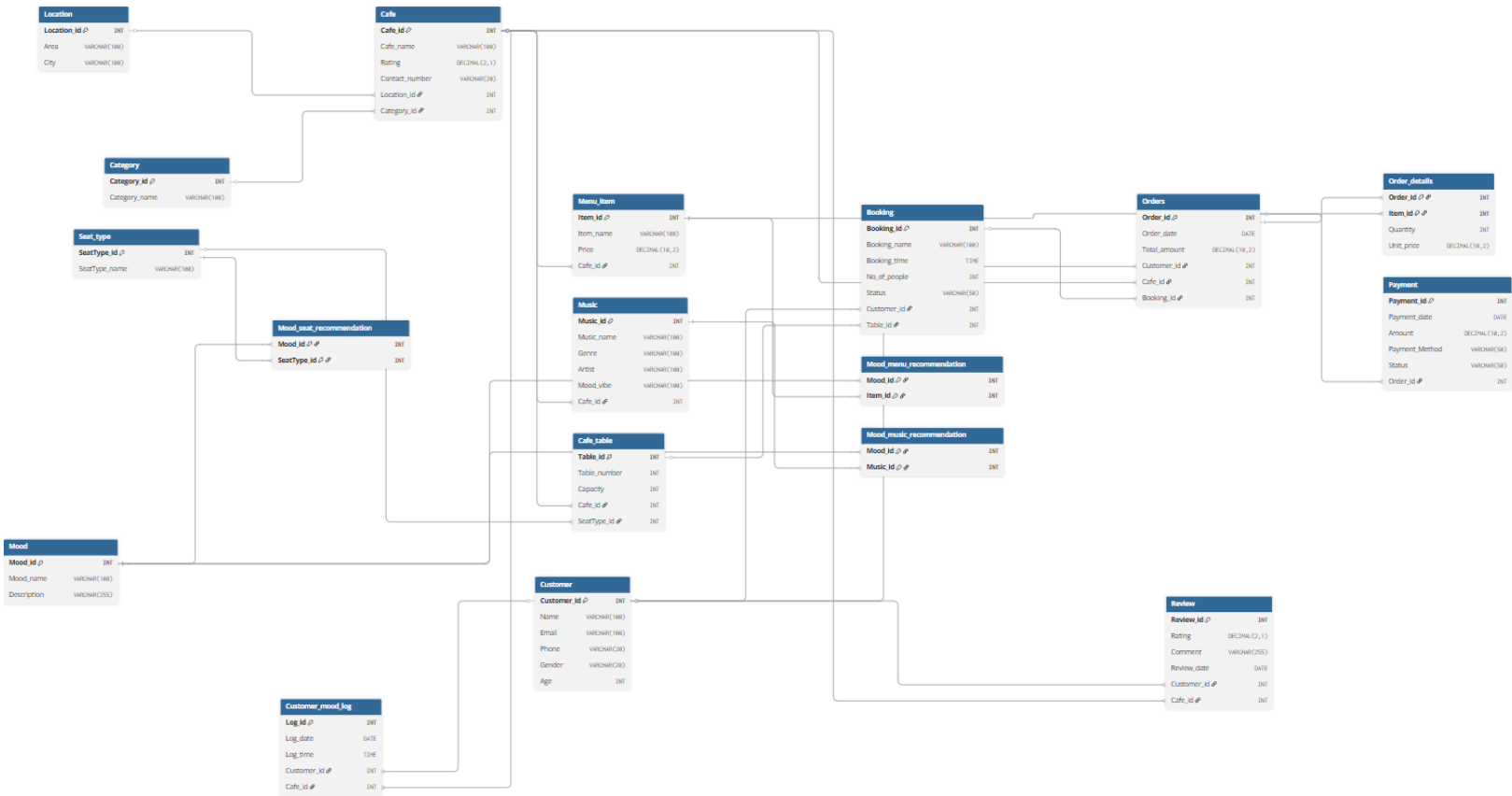
18. Mood_Seat_Recommendation

Description: This table stores mood-based seat recommendations. It connects moods with recommended seating types. The attributes 'Mood_ID' and 'SeatType_ID' together form the primary key and are foreign keys referencing the 'Mood' and 'Seat_Type' tables respectively.

Summary of Relationships :

The Mood-Based Café Recommendation System contains different types of relationships among its tables. Customers can make bookings, place orders, and give reviews. Cafés contain menu items, music, and seating arrangements. Moods are connected with menu items, music, and seat types through recommendation tables. These relationships help organize and manage customer activities, café services, bookings, payments, and mood-based recommendations efficiently within the database system.

3(iii). SCHEMA DIAGRAM



4.(a) Table Creation

```
-- 1. LOCATION TABLE
CREATE TABLE Location (
    Location_id INT PRIMARY KEY,
    Area VARCHAR(100),
    City VARCHAR(100)
);

-- 2. CATEGORY TABLE
CREATE TABLE Category (
    Category_id INT PRIMARY KEY,
    Category_name VARCHAR(100)
);

-- 3. CUSTOMER TABLE
CREATE TABLE Customer (
    Customer_id INT PRIMARY KEY,
    Name VARCHAR(100),
    Email VARCHAR(100),
    Phone VARCHAR(20),
    Gender VARCHAR(20),
    Age INT
);

-- 4. CAFE TABLE
CREATE TABLE Cafe (
    Cafe_id INT PRIMARY KEY,
    Cafe_name VARCHAR(100),
    Rating DECIMAL(2,1),
    Contact_number VARCHAR(20),
    Location_id INT,
    Category_id INT,
    FOREIGN KEY (Location_id) REFERENCES Location(Location_id),
    FOREIGN KEY (Category_id) REFERENCES Category(Category_id)
);

-- 5. MOOD TABLE
CREATE TABLE Mood (
    Mood_id INT PRIMARY KEY,
    Mood_name VARCHAR(100),
    Description VARCHAR(255)
);

-- 6. MENU_ITEM TABLE
CREATE TABLE Menu_item (
    Item_id INT PRIMARY KEY,
    Item_name VARCHAR(100),
    Price DECIMAL(10,2),
    Cafe_id INT,
    FOREIGN KEY (Cafe_id) REFERENCES Cafe(Cafe_id)
);

-- 7. MUSIC TABLE
CREATE TABLE Music (
    Music_id INT PRIMARY KEY,
    Music_name VARCHAR(100),
    Genre VARCHAR(100),
    Artist VARCHAR(100),
    Mood_vibe VARCHAR(100),
    Cafe_id INT,
    FOREIGN KEY (Cafe_id) REFERENCES Cafe(Cafe_id)
);

-- 8. SEAT_TYPE TABLE
CREATE TABLE Seat_type (
    SeatType_id INT PRIMARY KEY,
    SeatType_name VARCHAR(100)
);
```

```

-- 9. CAFE_TABLE TABLE
CREATE TABLE Cafe_table (
    Table_id INT PRIMARY KEY,
    Table_number INT,
    Capacity INT,
    Cafe_id INT,
    SeatType_id INT,
    FOREIGN KEY (Cafe_id) REFERENCES Cafe(Cafe_id),
    FOREIGN KEY (SeatType_id) REFERENCES Seat_type(SeatType_id)
);

-- 10. BOOKING TABLE
CREATE TABLE Booking (
    Booking_id INT PRIMARY KEY,
    Booking_name VARCHAR(100),
    Booking_time TIME,
    No_of_people INT,
    Status VARCHAR(50),
    Customer_id INT,
    Table_id INT,
    FOREIGN KEY (Customer_id) REFERENCES Customer(Customer_id),
    FOREIGN KEY (Table_id) REFERENCES Cafe_table(Table_id)
);

-- 11. ORDERS TABLE
CREATE TABLE Orders (
    Order_id INT PRIMARY KEY,
    Order_date DATE,
    Total_amount DECIMAL(10,2),
    Customer_id INT,
    Cafe_id INT,
    Booking_id INT,
    FOREIGN KEY (Customer_id) REFERENCES Customer(Customer_id),
    FOREIGN KEY (Cafe_id) REFERENCES Cafe(Cafe_id),
    FOREIGN KEY (Booking_id) REFERENCES Booking(Booking_id)
);

-- 12. ORDER_DETAILS TABLE
CREATE TABLE Order_details (
    Order_id INT,
    Item_id INT,
    Quantity INT,
    Unit_price DECIMAL(10,2),
    PRIMARY KEY (Order_id, Item_id),
    FOREIGN KEY (Order_id) REFERENCES Orders(Order_id),
    FOREIGN KEY (Item_id) REFERENCES Menu_item(Item_id)
);

-- 13. PAYMENT TABLE
CREATE TABLE Payment (
    Payment_id INT PRIMARY KEY,
    Payment_date DATE,
    Amount DECIMAL(10,2),
    Payment_Method VARCHAR(50),
    Status VARCHAR(50),
    Order_id INT,
    FOREIGN KEY (Order_id) REFERENCES Orders(Order_id)
);

-- 14. REVIEW TABLE
CREATE TABLE Review (
    Review_id INT PRIMARY KEY,
    Rating DECIMAL(2,1),
    Comment VARCHAR(255),
    Review_date DATE,
    Customer_id INT,
    Cafe_id INT,
    FOREIGN KEY (Customer_id) REFERENCES Customer(Customer_id),
    FOREIGN KEY (Cafe_id) REFERENCES Cafe(Cafe_id)
);

```

```

-- 15. CUSTOMER_MOOD_LOG TABLE
CREATE TABLE Customer_mood_log (
    Log_id INT PRIMARY KEY,
    Log_date DATE,
    Log_time TIME,
    Customer_id INT,
    Cafe_id INT,
    FOREIGN KEY (Customer_id) REFERENCES Customer(Customer_id),
    FOREIGN KEY (Cafe_id) REFERENCES Cafe(Cafe_id)
);

-- 16. MOOD_MENU_RECOMMENDATION TABLE
CREATE TABLE Mood_menu_recommendation (
    Mood_id INT,
    Item_id INT,
    PRIMARY KEY (Mood_id, Item_id),
    FOREIGN KEY (Mood_id) REFERENCES Mood(Mood_id),
    FOREIGN KEY (Item_id) REFERENCES Menu_item(Item_id)
);

-- 17. MOOD_MUSIC_RECOMMENDATION TABLE
CREATE TABLE Mood_music_recommendation (
    Mood_id INT,
    Music_id INT,
    PRIMARY KEY (Mood_id, Music_id),
    FOREIGN KEY (Mood_id) REFERENCES Mood(Mood_id),
    FOREIGN KEY (Music_id) REFERENCES Music(Music_id)
);

-- 18. MOOD_SEAT_RECOMMENDATION TABLE
CREATE TABLE Mood_seat_recommendation (
    Mood_id INT,
    SeatType_id INT,
    PRIMARY KEY (Mood_id, SeatType_id),
    FOREIGN KEY (Mood_id) REFERENCES Mood(Mood_id),
    FOREIGN KEY (SeatType_id) REFERENCES Seat_type(SeatType_id)
);

```

4.(b) Table information

1. Location Table

- **Location_id** – Primary key, uniquely identifies each location
- **Area** – Name of the area/location
- **City** – Name of the city

2. Category Table

- **Category_id** – Primary key, uniquely identifies each category
- **Category_name** – Name of cafe category

3. Customer Table

- **Customer_id** – Primary key, uniquely identifies each customer
- **Name** – Customer name
- **Email** – Customer email address
- **Phone** – Customer phone number
- **Gender** – Customer gender
- **Age** – Customer age

4. Cafe Table

- **Cafe_id** – Primary key, uniquely identifies each cafe
- **Cafe_name** – Name of the cafe
- **Rating** – Rating of the cafe
- **Contact_number** – Contact number of the cafe
- **Location_id** – Foreign key referencing Location table
- **Category_id** – Foreign key referencing Category table

5. Mood Table

- **Mood_id** – Primary key, uniquely identifies each mood
- **Mood_name** – Name of the mood
- **Description** – Description of the mood

6. Menu_item Table

- **Item_id** – Primary key, uniquely identifies each menu item
- **Item_name** – Name of the menu item
- **Price** – Price of the item
- **Cafe_id** – Foreign key referencing Cafe table

7. Music Table

- **Music_id** – Primary key, uniquely identifies each music item
- **Music_name** – Name of the music
- **Genre** – Genre/category of music
- **Artist** – Name of the artist
- **Mood_vibe** – Mood vibe of the music
- **Cafe_id** – Foreign key referencing Cafe table

8. Seat_type Table

- **SeatType_id** – Primary key, uniquely identifies seat type
- **SeatType_name** – Name/type of seat

9. Cafe_table Table

- **Table_id** – Primary key, uniquely identifies each table
- **Table_number** – Table number of the cafe
- **Capacity** – Number of seats available
- **Cafe_id** – Foreign key referencing Cafe table
- **SeatType_id** – Foreign key referencing Seat_type table

10. Booking Table

- **Booking_id** – Primary key, uniquely identifies each booking
- **Booking_name** – Name of the booking
- **Booking_time** – Booking time
- **No_of_people** – Number of people
- **Status** – Booking status

- **Customer_id** – Foreign key referencing Customer table
- **Table_id** – Foreign key referencing Cafe_table table

11. Orders Table

- **Order_id** – Primary key, uniquely identifies each order
- **Order_date** – Date of order
- **Total_amount** – Total amount of order
- **Customer_id** – Foreign key referencing Customer table
- **Cafe_id** – Foreign key referencing Cafe table
- **Booking_id** – Foreign key referencing Booking table

12. Order_details Table

- **Order_id** – Foreign key referencing Orders table
- **Item_id** – Foreign key referencing Menu_item table
- **Quantity** – Quantity of ordered items
- **Unit_price** – Price per unit item
- **Primary Key** – Combination of Order_id and Item_id

13. Payment Table

- **Payment_id** – Primary key, uniquely identifies payment
- **Payment_date** – Date of payment
- **Amount** – Payment amount
- **Payment_Method** – Payment method
- **Status** – Payment status
- **Order_id** – Foreign key referencing Orders table

14. Review Table

- **Review_id** – Primary key, uniquely identifies review
- **Rating** – Customer rating
- **Comment** – Customer comment/review
- **Review_date** – Review date
- **Customer_id** – Foreign key referencing Customer table
- **Cafe_id** – Foreign key referencing Cafe table

15. Customer_mood_log Table

- **Log_id** – Primary key, uniquely identifies each log
- **Log_date** – Date of mood log
- **Log_time** – Time of mood log
- **Customer_id** – Foreign key referencing Customer table
- **Cafe_id** – Foreign key referencing Cafe table

17. Mood_music_recommendation Table

- **Mood_id** – Foreign key referencing Mood table
- **Music_id** – Foreign key referencing Music table
- **Primary Key** – Combination of Mood_id and Music_id

16. Mood_menu_recommendation Table

- **Mood_id** – Foreign key referencing Mood table
- **Item_id** – Foreign key referencing Menu_item table
- **Primary Key** – Combination of Mood_id and Item_id

18. Mood_seat_recommendation Table

- **Mood_id** – Foreign key referencing Mood table
- **SeatType_id** – Foreign key referencing Seat_type table
- **Primary Key** – Combination of Mood_id and SeatType_id

5. Data Insertion

-- 1. LOCATION TABLE

```
INSERT INTO Location (Location_id, Area, City) VALUES
```

```
(1, 'Dhanmondi', 'Dhaka'),  
(2, 'Banani', 'Dhaka'),  
(3, 'Gulshan', 'Dhaka'),  
(4, 'Mirpur', 'Dhaka'),  
(5, 'Uttara', 'Dhaka');
```

```
SELECT * FROM [dbo].[Location];
```

-- 2. CATEGORY TABLE

```
INSERT INTO Category (Category_id, Category_name) VALUES
```

```
(1, 'Romantic Cafe'),  
(2, 'Study Cafe'),  
(3, 'Luxury Cafe'),  
(4, 'Couple Cafe'),  
(5, 'Theme Cafe');
```

```
SELECT * FROM [dbo].[Category];
```

-- 3. CUSTOMER TABLE

```
INSERT INTO Customer (Customer_id, Name, Email, Phone, Gender, Age) VALUES
```

```
(1, 'Noshin', 'noshin@gmail.com', '01711111111', 'Female', 21),  
(2, 'Jyoti', 'jyoti@gmail.com', '01822222222', 'Female', 22),  
(3, 'Lamia', 'lamia@gmail.com', '01555555555', 'Female', 21),  
(4, 'Ariana', 'ariana@gmail.com', '01666666666', 'Female', 23),  
(5, 'Muntasir', 'muntasir@gmail.com', '01999999999', 'Male', 24);
```

```
SELECT * FROM [dbo].[Customer];
```

-- 4. CAFE TABLE

```
INSERT INTO Cafe (Cafe_id, Cafe_name, Rating, Contact_number, Location_id, Category_id) VALUES
```

```
(1, 'Cafe Mocha', 4.5, '01788888888', 1, 1),  
(2, 'Coffee House', 4.2, '01877777777', 2, 2),  
(3, 'Moonlight Cafe', 4.8, '01544444444', 3, 3),  
(4, 'Sky Lounge', 4.6, '01677777777', 4, 4),  
(5, 'Dream Cafe', 4.7, '01988888888', 5, 5);
```

```
SELECT * FROM [dbo].[Cafe];
```

-- 5. MOOD TABLE

```
INSERT INTO Mood (Mood_id, Mood_name, Description) VALUES
```

```
(1, 'Happy', 'Feeling joyful'),  
(2, 'Relaxed', 'Feeling calm'),  
(3, 'Energetic', 'Feeling active'),  
(4, 'Romantic', 'Feeling loved'),  
(5, 'Sad', 'Feeling emotional');
```

```
SELECT * FROM [dbo].[Mood];
```

```

-- 6. MENU_ITEM TABLE
INSERT INTO Menu_item (Item_id, Item_name, Price, Cafe_id) VALUES
(1, 'Cold Coffee', 250, 1),
(2, 'Cappuccino', 300, 2),
(3, 'Brownie', 200, 3),
(4, 'Cheese Cake', 350, 4),
(5, 'Chicken Pasta', 450, 5);

SELECT * FROM [dbo].[Menu_item];

-- 7. MUSIC TABLE
INSERT INTO Music (Music_id, Music_name, Genre, Artist, Mood_vibe, Cafe_id) VALUES
(1, 'Perfect', 'Pop', 'Ed Sheeran', 'Romantic', 1),
(2, 'Night Changes', 'Pop', 'One Direction', 'Relaxed', 2),
(3, 'Love Story', 'Pop', 'Taylor Swift', 'Happy', 3),
(4, 'Until I Found You', 'Romantic', 'Stephen Sanchez', 'Romantic', 4),
(5, 'Someone Like You', 'Soft Pop', 'Adele', 'Sad', 5);

SELECT * FROM [dbo].[Music];

-- 8. SEAT_TYPE TABLE
INSERT INTO Seat_type (SeatType_id, SeatType_name) VALUES
(1, 'Couple Seat'),
(2, 'Window Seat'),
(3, 'Outdoor Seat'),
(4, 'Private Booth'),
(5, 'Rooftop Seat');

SELECT * FROM [dbo].[Seat_type];

-- 9. CAFE_TABLE TABLE
INSERT INTO Cafe_table (Table_id, Table_number, Capacity, Cafe_id, SeatType_id) VALUES
(1, 101, 4, 1, 1),
(2, 102, 2, 2, 2),
(3, 105, 2, 3, 3),
(4, 110, 4, 4, 4),
(5, 120, 6, 5, 5);

SELECT * FROM [dbo].[Cafe_table];

-- 10. BOOKING TABLE
INSERT INTO Booking (Booking_id, Booking_name, Booking_time, No_of_people, Status, Customer_id,
Table_id) VALUES
(1, 'Noshin', '18:00:00', 2, 'Confirmed', 1, 1),
(2, 'Jyoti', '15:00:00', 3, 'Pending', 2, 2),
(3, 'Lamia', '20:00:00', 4, 'Confirmed', 3, 3),
(4, 'Tabassum', '17:00:00', 5, 'Confirmed', 4, 4),
(5, 'John', '21:00:00', 2, 'Pending', 5, 5);

SELECT * FROM [dbo].[Booking];

-- 11. ORDERS TABLE
INSERT INTO Orders (Order_id, Order_date, Total_amount, Customer_id, Cafe_id, Booking_id) VALUES
(1, '2026-05-05', 500, 1, 1, 1),
(2, '2026-05-06', 700, 2, 2, 2),
(3, '2026-05-09', 650, 3, 3, 3),
(4, '2026-05-10', 850, 4, 4, 4),
(5, '2026-05-11', 950, 5, 5, 5);

SELECT * FROM [dbo].[Orders];

-- 12. ORDER_DETAILS TABLE
INSERT INTO Order_details (Order_id, Item_id, Quantity, Unit_price) VALUES
(1, 1, 2, 250),
(2, 2, 2, 350),
(3, 3, 3, 300),
(4, 4, 2, 425),
(5, 5, 2, 475);

```

```

SELECT * FROM [dbo].[Order_details];

-- 13. PAYMENT TABLE
INSERT INTO Payment (Payment_id, Payment_date, Amount, Payment_Method, Status, Order_id) VALUES
(1, '2026-05-05', 500, 'Cash', 'Paid', 1),
(2, '2026-05-06', 700, 'Bkash', 'Paid', 2),
(3, '2026-05-09', 650, 'Nagad', 'Paid', 3),
(4, '2026-05-10', 850, 'Card', 'Paid', 4),
(5, '2026-05-11', 950, 'Bkash', 'Paid', 5);

SELECT * FROM [dbo].[Payment];

-- 14. REVIEW TABLE
INSERT INTO Review (Review_id, Rating, Comment, Review_date, Customer_id, Cafe_id) VALUES
(1, 5, 'Excellent environment', '2026-05-05', 1, 1),
(2, 4, 'Nice music', '2026-05-06', 2, 2),
(3, 5, 'Very comfortable', '2026-05-07', 3, 3),
(4, 5, 'Amazing rooftop view', '2026-05-10', 4, 4),
(5, 4, 'Very peaceful place', '2026-05-11', 5, 5);

SELECT * FROM [dbo].[Review];

-- 15. CUSTOMER_MOOD_LOG TABLE
INSERT INTO Customer_mood_log (Log_id, Log_date, Log_time, Customer_id, Cafe_id) VALUES
(1, '2026-05-05', '18:00:00', 1, 1),
(2, '2026-05-06', '15:00:00', 2, 2),
(3, '2026-05-07', '19:00:00', 3, 3),
(4, '2026-05-10', '17:00:00', 4, 4),
(5, '2026-05-11', '21:00:00', 5, 5);

SELECT * FROM [dbo].[Customer_mood_log];

-- 16. MOOD_MENU_RECOMMENDATION TABLE
INSERT INTO Mood_menu_recommendation (Mood_id, Item_id) VALUES
(1, 1),
(2, 2),
(3, 3),
(4, 4),
(5, 5);

SELECT * FROM [dbo].[Mood_menu_recommendation];

-- 17. MOOD_MUSIC_RECOMMENDATION TABLE
INSERT INTO Mood_music_recommendation (Mood_id, Music_id) VALUES
(1, 1),
(2, 2),
(3, 3),
(4, 4),
(5, 5);

SELECT * FROM [dbo].[Mood_music_recommendation];

-- 18. MOOD_SEAT_RECOMMENDATION TABLE
INSERT INTO Mood_seat_recommendation (Mood_id, SeatType_id) VALUES
(1, 1),
(2, 2),
(3, 3),
(4, 4),
(5, 5);

SELECT * FROM [dbo].[Mood_seat_recommendation];

```

6. Queries (Question, DML Query, Output Screenshot)

1. Simple SELECT Query

Question: Show café name

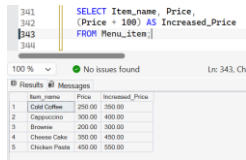
```
SELECT Cafe_name  
FROM Cafe;
```

	Cafe_name
1	Cafe Mocha
2	Coffee House
3	Moonlight Cafe
4	Sky Lounge
5	Dream Cafe

2. Arithmetic Expression

Question: Increase menu item price by 50.

```
SELECT Item_name,  
Price + 50 AS New_Price  
FROM Menu_item;
```



The screenshot shows a SQL query editor with the following query:

```
SELECT Item_name, Price,  
(Price + 100) AS Increased_Price  
FROM Menu_item;
```

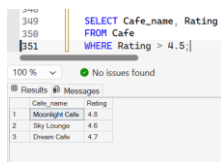
The results pane shows the following output:

Item_name	Price	Increased_Price
1. Cold Coffee	250.00	350.00
2. Cappuccino	300.00	400.00
3. Brownie	200.00	300.00
4. Cheese Cake	300.00	400.00
5. Chicken Pasta	400.00	500.00

3. WHERE Clause

Question: Show cafés with rating above 4

```
SELECT Cafe_name, Rating  
FROM Café  
WHERE Rating > 4;
```



The screenshot shows a SQL query editor with the following query:

```
SELECT Cafe_name, Rating  
FROM Café  
WHERE Rating > 4.5;
```

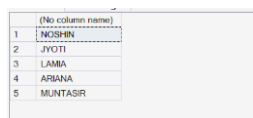
The results pane shows the following output:

Cafe_name	Rating
1. Moonlight Cafe	4.8
2. Sky Lounge	4.6
3. Dream Cafe	4.7

4. String Function

Question: Show customer names in uppercase

```
SELECT UPPER(Name)  
FROM Customer;
```



The screenshot shows a SQL query editor with the following query:

```
SELECT UPPER(Name)  
FROM Customer;
```

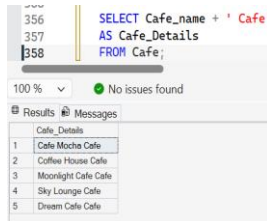
The results pane shows the following output:

(No column name)
1. NOSHIN
2. JYOTI
3. LAMIA
4. ARIANA
5. MUNTASIR

5. String Concatenation

Question: Show café details

```
SELECT Cafe_name + ' Cafe'
AS Cafe_Details
FROM Cafe;
```



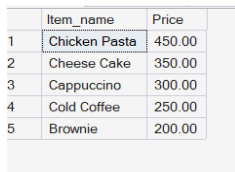
The screenshot shows a SQL query editor with the query: `SELECT Cafe_name + ' Cafe' AS Cafe_Details FROM Cafe;`. Below the editor, the results are displayed in a table with 5 rows and 1 column.

	Cafe_Details
1	Cafe Mocha Cafe
2	Coffee House Cafe
3	Moonlight Cafe Cafe
4	Sky Lounge Cafe
5	Dream Cafe Cafe

6. Order By

Question: Show menu items by highest price

```
SELECT Item_name, Price
FROM Menu_item
ORDER BY Price DESC;
```



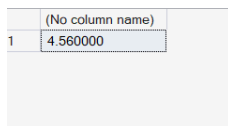
The screenshot shows a SQL query editor with the query: `SELECT Item_name, Price FROM Menu_item ORDER BY Price DESC;`. Below the editor, the results are displayed in a table with 5 rows and 2 columns.

	Item_name	Price
1	Chicken Pasta	450.00
2	Cheese Cake	350.00
3	Cappuccino	300.00
4	Cold Coffee	250.00
5	Brownie	200.00

7. Avg()

Question: Find average café rating

```
SELECT AVG(Rating)
FROM Cafe;
```



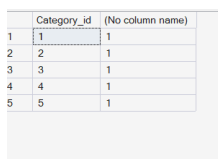
The screenshot shows a SQL query editor with the query: `SELECT AVG(Rating) FROM Cafe;`. Below the editor, the results are displayed in a table with 1 row and 1 column.

	(No column name)
1	4.560000

8. Group By Query

Question: Count cafés in each category

```
SELECT Category_id,
COUNT(*)
FROM Café
GROUP BY Category_id;
```



The screenshot shows a SQL query editor with the query: `SELECT Category_id, COUNT(*) FROM Café GROUP BY Category_id;`. Below the editor, the results are displayed in a table with 5 rows and 2 columns.

	Category_id	(No column name)
1	1	1
2	2	1
3	3	1
4	4	1
5	5	1

9. Join Query

Question: Show café names with city

```
SELECT Cafe_name, City
FROM Café
JOIN Location
ON Cafe.Location_id = Location.Location_id;
```

	Cafe_name	City
1	Cafe Mocha	Chaska
2	Coffee House	Chaska
3	Moonlight Cafe	Chaska
4	Sky Lounge	Chaska
5	Dream Cafe	Chaska

10. In Query

Question: Show confirmed bookings

```
SELECT *
FROM Booking
WHERE Status IN ('Confirmed');
```

Booking_id	Booking_name	Booking_time	No_of_people	Status	Customer_id	Table_id
1	Evening Booking	18:00:00:0000000	2	Confirmed	1	1
2	Dinner Plan	20:00:00:0000000	4	Confirmed	3	3

11. Like Query

Question: Find music names containing Love

```
SELECT *
FROM Music
WHERE Music_name LIKE '%Love%';
```

	Music_id	Music_name	Genre	Artist	Mood_vibe	Cafe_id
1	3	Love Story	Pop	Taylor Swift	Happy	3

12. Subquery

Question: Show café names having rating greater than the average rating

```
SELECT Cafe_name, Rating
FROM Cafe
WHERE Rating >
(
SELECT AVG(Rating)
FROM Café
);
```

	Cafe_name
1	Cafe Mocha
2	Coffee House
3	Moonlight Cafe

13. View Query

Question: Create a café rating view

```
CREATE VIEW vw_Cafe
AS
SELECT Cafe_name, Rating
FROM Cafe;
```

	Cafe_name	Rating
1	Cafe Mocha	4.5
2	Coffee House	4.2
3	Moonlight Cafe	4.8
4	Sky Lounge	4.6
5	Dream Cafe	4.7

14. Trigger Query

Question: Create trigger after insert

```
CREATE TRIGGER trg_Test
ON Payment
AFTER INSERT
AS
BEGIN
PRINT 'Payment Added';
END;
INSERT INTO Payment
VALUES (6, '2026-05-16', 500, 'Cash', 'Paid', 2);
```

```
Payment Added

(1 row affected)

Completion time: 2026-05-16T22:50:10.7813358+06:00
```

15. Update Query

Question: Update café rating

```
UPDATE Cafe
SET Rating = 4.9
WHERE Cafe_id = 1;
```

	Cafe_id	Cafe_name	Rating	Contact_number	Location_id	Category_id
1	1	Cafe Mocha	4.9	01780808888	1	1
2	2	Coffee House	4.2	01677777777	2	2
3	3	Moonlight Cafe	4.8	01544444444	3	3
4	4	Sky Lounge	4.6	01677777777	4	4
5	5	Dream Cafe	4.7	01908080888	5	5

16. Delete Query

Question: Delete a review

```
DELETE FROM Review
WHERE Review_id = 5;
```

	Review_id	Rating	Comment	Review_date	Customer_id	Cafe_id
1	1	5.0	Excellent environment	2026-05-05	1	1
2	2	4.0	Nice music	2026-05-06	2	2
3	3	5.0	Very comfortable	2026-05-07	3	3

7. Mapping with CEP(Complex Engineering Problem) Attributes

K = Knowledge

P = Problem Solving

A = Engineering Activities

Code	Attribute	How Your Project Reflects It
K2	Engineering Mathematics, numerical analysis	Applied fundamental DBMS concepts such as entity relationships (1–M, M–M), primary keys, foreign keys, and referential integrity in designing the Mood-Based Café Recommendation System.
K3	Engineering fundamentals	The database schema follows software engineering fundamentals such as referential integrity, entity relationships, modularity, joins, views, triggers, and stored procedures.
K5	Engineering design	The schema design supports real-world café management, booking systems, customer mood tracking, recommendation systems, and payment management.
K6	Engineering practice (technology)	The project was implemented using Microsoft SQL Server and practical database technologies including SQL DDL, DML, triggers, views, stored procedures, and relational database management.

Code	Attributes	How Your Project Reflects It
P1	Depth of Knowledge required	The project involves relational database modeling, schema design, triggers, stored procedures, normalization, and relationship management.
P2	Conflicting technical, engineering, or other requirements	Manages multiple interconnected tables while maintaining referential integrity.
P3	Requires database modeling and system design	Transforms real-world café and mood systems into database entities and relationships.
P6	Supports different users	Supports customers, café owners, managers, and administrators.
P7	High-level problem with many subcomponents	Contains multiple interconnected tables, recommendations, bookings, and payments.

Code	Attributes	How Your Project Reflects It
A1	Uses different database resources	Manages cafés, customers, bookings, menu items, payments, and recommendations.
A2	Requires resolution of complex interactions	Handles relationships among customers, cafés, bookings, moods, and payments.
A3	Requires innovative application of knowledge	Uses triggers, views, joins, and stored procedures for automation.
A4	Significant consequences for society & environment	Improves digital café management and customer experience.

